

Notes: Dessein (RES, 2002)

Authority and Communication in Organizations

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Model primitives and measurement	4
3.1	Actors and timing	4
3.2	Preferences and bias	4
3.3	Information structure	4
3.4	Contracts and noncontractibility	4
4	Models and theoretical specifications	5
4.1	Delegation	5
4.2	Communication as cheap talk	5
4.3	Crawford–Sobel partition condition	5
4.4	Uniform distribution and quadratic intuition	5
4.5	Delegation versus communication under a uniform prior	6
4.6	Small-bias result	6
4.7	When communication can dominate	6
4.8	Intermediary delegation	6
4.9	Delegation with veto power	6
5	Identification and theoretical design	7
5.1	What the paper is identifying theoretically	7
5.2	Counterfactual regimes	7
5.3	Key assumptions supporting the design	7
5.4	Why the comparison is not obvious	7
5.5	Main theoretical threat and response	8
6	Tables and figures	8
6.1	Figure 1: Partition Elements Are Increasingly Large	8
6.2	Figure 2: Given b and $F(m)$, Communication Is Informative and Dominates Delegation . . .	9
6.3	Figure 3: Delegation Decision in the Leading Example Crawford and Sobel (1982)	9
7	Short summary	10
8	Conclusion	10
8.1	Core conclusion	10
8.2	Economic intuition	10

8.3	Organizational implications	10
8.4	Policy and management takeaway	11
8.5	Limitations	11
8.6	Takeaway	11

1 Big picture

- **Question.** When should an uninformed principal delegate decision rights to a better-informed but biased agent, rather than keep authority and ask the agent to communicate his information?
- **Core trade-off.** Delegation preserves information but loses control; communication preserves control but strategically garbles information.
- **Main result.** Delegation is often optimal when the agent's bias is small relative to the principal's uncertainty about the environment.
- **Surprise.** Keeping authority and communicating does not generally dominate delegation, even though the principal could in principle rubber-stamp the agent's proposal.
- **Mechanism.** When preferences differ, the agent exaggerates in the direction of his bias. Equilibrium communication must therefore become noisy, especially for messages recommending actions in the direction of the bias.
- **Uniform benchmark.** With a uniform prior over the state, the principal prefers delegation whenever informative communication is feasible.
- **Small-bias theorem.** For any smooth distribution, as the bias tends to zero, communication generates decision errors that are infinitely large relative to the delegation error.
- **Exception.** If the principal has a very precise prior, communication can dominate delegation because the uninformed action is already close to the optimum.
- **Intermediaries.** For moderate biases, the principal may delegate to an intermediary whose preferences lie between hers and the agent's.
- **Veto rights.** Retaining veto power can reduce extreme mistakes, but it often makes the principal worse off because it adds variance to the implemented bias.
- **Economic takeaway.** "Knowledge is power" because authority can be a substitute for costly strategic communication. Delegation is not only an incentive device; it is also an information-preservation device.

2 Incremental contribution of the paper

- **Delegation versus cheap talk in one framework.** The paper embeds Crawford–Sobel communication in an incomplete-contracting allocation-of-authority problem. This makes communication and delegation directly comparable.
- **Purely informational theory of authority.** Earlier work often emphasized authority because it changes incentives to acquire information. Dessein instead holds the information structure fixed and shows that authority matters because it changes how information is used.
- **Delegation can dominate communication even for small conflicts.** The paper overturns the intuition that small preference differences make communication almost costless. In equilibrium, fine communication requires increasingly coarse partitions relative to the bias.
- **Information loss versus control loss.** The paper formalizes a clean organizational trade-off: delegation creates a predictable bias of size b , while communication creates noisy decision rules whose error can exceed that bias.
- **Intermediary as a commitment device.** The paper shows how delegating to an intermediary can be optimal because it limits the agent's discretion without returning to fully strategic communication with the principal.
- **Veto rights need not help.** A veto option sounds like a weak improvement over full delegation. The paper shows that it can reduce expected utility unless the conflict is extreme.
- **Organizational implication.** Decentralization is not only about empowering local managers or motivating information acquisition. It can be the optimal way to avoid strategic distortion of soft information.

3 Model primitives and measurement

3.1 Actors and timing

There is a principal, an agent, one project to choose, and a one-dimensional action space $y \in \mathbb{R}$. The agent observes the relevant state m ; the principal does not. The principal initially controls critical resources needed to implement the project.

The timing is:

1. The principal decides whether to delegate authority over the critical resources.
2. The agent observes m .
3. If authority is delegated, the agent implements his preferred project. If authority is retained, the principal may ask for a recommendation and then chooses the project.
4. The agent implements the chosen project.

Interpretation. Authority is modeled as formal control over critical resources. The agent's information is useful but soft: it cannot be certified or written into a contract.

3.2 Preferences and bias

The principal's ideal action is $y = m$. Her payoff can be written as

$$U_P(y, m) = U_P(m, m) - \ell(|y - m|),$$

where $\ell'(\cdot) > 0$ and $\ell'(0) = 0$.

The agent's ideal action is $y = m + b$:

$$U_A(y, m, b) = U_A(m + b, m, b) - \ell_A(|y - (m + b)|),$$

where $b > 0$ is the systematic preference divergence, or bias.

Interpretation. The bias captures predictable organizational conflicts: empire building, short-termism, risk aversion, status-quo bias, excessive concern for subordinates, or project-size preferences.

3.3 Information structure

Only the agent observes m . The principal knows that

$$m \sim F(m), \quad m \in [-L, L],$$

where F is twice differentiable with density f .

Interpretation. The agent's informational advantage comes from proximity to customers, suppliers, technology, or local market conditions. The principal's problem is how to use this information without letting bias distort the project choice too much.

3.4 Contracts and noncontractibility

Projects cannot be contracted upon. The principal cannot write a mechanism specifying the project as a function of the agent's report. However, the principal can allocate authority over the resources required to initiate a project.

Interpretation. This is why the model is about authority rather than monetary transfers. The principal can choose who has decision rights, but cannot contract directly on the decision.

4 Models and theoretical specifications

4.1 Delegation

If the principal delegates authority, the agent chooses

$$y = m + b.$$

The principal's payoff is

$$U_P(m + b, m) = U_P(m, m) - \ell(b).$$

The principal delegates rather than takes an uninformed action if b is below a cutoff b' satisfying

$$\ell(b') = \min_y \int_{-L}^L \ell(|y - m|) dF(m).$$

Interpretation. Delegation creates a predictable loss of control: the action is always shifted by b . But it fully uses the agent's information about m .

4.2 Communication as cheap talk

If the principal retains authority, the agent sends a message and the principal chooses an action. A communication equilibrium is characterized by a partition

$$-L = a_0 < a_1 < \dots < a_N = L.$$

For each interval (a_{i-1}, a_i) , the agent sends a message that induces the principal to choose

$$y(a_{i-1}, a_i) = \arg \max_y \int_{a_{i-1}}^{a_i} U_P(y, m) f(m) dm.$$

Interpretation. Communication is informative only if the partition has more than one element. Because the agent has a bias, messages cannot reveal the exact state; they only reveal which interval contains the state.

4.3 Crawford–Sobel partition condition

At a boundary a_i , the agent must be indifferent between the two actions induced by the adjacent messages. The equilibrium partition therefore satisfies an arbitrage condition of the form

$$U_A(y(a_{i-1}, a_i), a_i, b) = U_A(y(a_i, a_{i+1}), a_i, b).$$

Interpretation. This indifference condition is the screening mechanism. Partition sizes adjust so that the agent does not want to exaggerate the state by sending a higher message.

4.4 Uniform distribution and quadratic intuition

When F is uniform, the principal's action after message interval (a_{i-1}, a_i) is the midpoint. The indifference condition implies

$$a_{i+1} - a_i = a_i - a_{i-1} + 4b.$$

Lemma 1. For a uniform distribution, each successive partition element is $4b$ larger than the preceding one. Messages that recommend larger actions must be noisier to keep the biased agent truthful.

Interpretation. This is the core mathematical force behind the paper. Good communication requires many messages, but many messages imply partition elements that become large relative to the bias. Therefore, communication becomes unattractive precisely when it looks informative in absolute terms.

4.5 Delegation versus communication under a uniform prior

Proposition 2. If $F(m)$ is uniformly distributed over $[-L, L]$, the principal prefers delegation to communication whenever the bias is such that informative communication is feasible.

Interpretation. The proposition says that in the benchmark Crawford–Sobel environment, once communication can transmit any information, the principal is still better off delegating. The loss of control under delegation is smaller than the endogenous information loss under cheap talk.

4.6 Small-bias result

Proposition 3. For any smooth distribution F , in the most informative communication equilibrium,

$$\lim_{b \rightarrow 0} \frac{E(|y^* - m|)}{b} = \infty.$$

Interpretation. As the agent’s bias becomes small, delegation error shrinks one-for-one with b . Communication error shrinks more slowly because the partition structure still requires noisy messages. This explains why small conflicts can still justify delegation.

4.7 When communication can dominate

If the principal’s prior is very precise, an uninformed decision may already be close to optimal. For symmetric F and $b < L/2$, informative communication dominates delegation if roughly

$$\int_{-L}^L \ell(|m|) dF(m) \leq \ell(b).$$

Interpretation. Communication can beat delegation when uncertainty is small relative to bias. In that case, preserving control is valuable because the principal does not need much information to choose a reasonably good action.

4.8 Intermediary delegation

The paper then allows delegation to an intermediary with bias between the principal’s and the agent’s. In the leading Crawford–Sobel example with uniform F on $[0, 1]$ and quadratic loss:

Proposition 5. If $b \leq 1/6$, the principal delegates to the agent. If $b \in (1/6, \sqrt{2}/4)$, she delegates to an intermediary. If $b > \sqrt{2}/4$, there is neither delegation nor informative communication.

Interpretation. The intermediary is not merely a messenger. Delegating to the intermediary commits the organization to restrict the agent’s discretion while avoiding the full information loss of communication with the principal.

4.9 Delegation with veto power

Under the closed rule, the agent proposes a project and the principal can accept it or veto it in favor of a status quo y_0 .

Propositions 6–7. With a central status quo in the leading example, pure delegation strictly dominates delegation with veto power for small and moderate biases. Veto power becomes useful only when the bias is large enough.

Interpretation. A veto right sounds like a weak improvement over full delegation, but it can make the principal worse off. It creates variation in the implemented bias without reducing the average bias enough, unless the agent's conflict is extreme.

5 Identification and theoretical design

5.1 What the paper is identifying theoretically

The paper is not an empirical identification design. It is a theoretical comparison of organizational regimes. The object being identified is the condition under which a principal should allocate formal authority to an informed but biased agent rather than keep authority and rely on communication.

Interpretation. The paper isolates the informational role of authority. It asks what happens to decision quality when authority changes, holding the agent's information and bias fixed.

5.2 Counterfactual regimes

The model compares the following regimes:

- **Uninformed principal control:** the principal chooses without useful communication.
- **Communication with principal control:** the agent sends a cheap-talk message and the principal chooses.
- **Full delegation:** the agent chooses the action after observing the state.
- **Delegation to an intermediary:** a partially biased intermediary receives authority.
- **Delegation with veto:** the agent proposes and the principal can veto.

Interpretation. These regimes are counterfactual governance structures. The paper's claims come from comparing expected utilities across them under the same primitives.

5.3 Key assumptions supporting the design

- **Soft information:** the agent cannot certify the state.
- **Noncontractible projects:** the principal cannot write a revelation mechanism over actions.
- **Exogenous bias:** the conflict b is treated as a primitive in the core model.
- **Formal authority over critical resources:** the principal can delegate or retain control.
- **No reversal after implementation:** once the project is initiated, the action cannot be undone.

Interpretation. The design deliberately removes monetary transfers and hard evidence to focus on allocation of authority as the main organizational instrument.

5.4 Why the comparison is not obvious

A naive argument says that retaining authority should dominate delegation because the principal can always rubber-stamp the agent's recommendation. The paper shows why this logic fails: the agent anticipates correction and strategically garbles communication.

Interpretation. The cost of principal control is not just delay or bureaucracy. It is endogenous loss of information caused by the agent's incentive to manipulate soft information.

5.5 Main theoretical threat and response

The strongest threat is that the result might depend on a special uniform distribution. The paper addresses this by proving the small-bias result for any smooth distribution and by studying steep priors as the main case in which communication can dominate.

Interpretation. The uniform model gives the sharpest result, but the broader lesson is not only a uniform-prior artifact. Delegation is robustly attractive when bias is small relative to uncertainty.

6 Tables and figures

6.1 Figure 1: Partition Elements Are Increasingly Large

Read:

- a_0, a_1, a_2 are adjacent partition boundaries in a cheap-talk equilibrium.
- Y_1 and Y_2 are the principal's actions induced by the two messages.
- At boundary a_1 , the agent must be indifferent between inducing Y_1 and inducing Y_2 .
- Because the agent wants actions shifted upward by b , the second interval must be wider.

Economic message. The figure explains why communication becomes costly. A biased agent can be kept honest only by making high-action messages less precise. Therefore, cheap talk loses information endogenously.

Detailed interpretation. If the agent recommends a large project, the principal knows that the recommendation may reflect bias rather than information. The equilibrium response is to make large recommendations correspond to wider state intervals. This disciplines exaggeration, but it also means the principal learns less exactly where the agent's bias is most tempting.

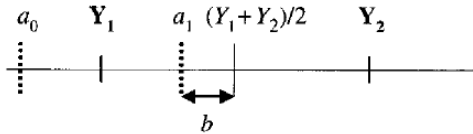


FIGURE 1

$$a_2 - a_1 = a_1 - a_0 + 4b$$

Figure 1: Partition elements are increasingly large. The figure illustrates the boundary condition that makes the agent indifferent between adjacent messages. Under the uniform-quadratic benchmark, the next partition interval must be $4b$ larger than the previous one. This visual is the mechanical heart of the paper: truthful communication requires larger and larger message intervals as recommendations move in the direction of the agent's bias.

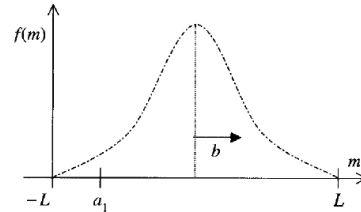


FIGURE 2

Given b and $F(m)$, communication is informative and dominates delegation

Figure 2: Given b and $F(m)$, communication is informative and dominates delegation. The figure shows a steep single-peaked prior over the state. When the prior is sufficiently precise, the principal can keep authority and communicate because her prior already pins down the desirable action fairly well. This is the main exception to the delegation result.

6.2 Figure 2: Given b and $F(m)$, Communication Is Informative and Dominates Delegation

Read:

- The horizontal axis is the state m on $[-L, L]$.
- The density $f(m)$ is steep and concentrated near the center.
- The bias b shifts the agent's ideal point away from the principal's ideal point.
- Communication dominates when the prior itself is sufficiently informative relative to the bias.

Economic message. Communication can dominate delegation when uncertainty is low. In that case, the principal gives up little information by retaining control, but avoids the systematic delegated bias.

Detailed interpretation. Figure 2 is important because it prevents an overstatement of the paper's result. Delegation is not always optimal. If the principal is already well informed through her prior, she can keep control and use even coarse communication to refine her decision. This is most plausible in stable environments or routine decisions where local information is not very valuable.

6.3 Figure 3: Delegation Decision in the Leading Example Crawford and Sobel (1982)

b	1/6	1/4	$\sqrt{3}/6$	$\sqrt{2}/4$
Authority Principal	COMM		NO COMM	
Authority Endogenous	DELEG			NO DELEG
Authority Endogenous, Intermediary	DELEG to A	DELEG to I		NO DELEG

FIGURE 3
Delegation decision in leading example Crawford and Sobel (1982)

Figure 3: Delegation decision in the leading example Crawford and Sobel (1982). The figure summarizes the optimal authority regime as the bias b rises. It compares principal authority, endogenous delegation, and endogenous delegation with an intermediary. The thresholds show how the optimal regime shifts from delegation to the agent, to delegation to an intermediary, to no delegation as preference conflict grows.

Read:

- The top row shows what happens when the principal keeps authority: communication is possible for low b , but becomes uninformative for high b .
- The middle row shows the pure delegation choice: delegation is optimal for low and intermediate b but not for large b .
- The bottom row adds a possible intermediary: the principal delegates to the agent for low b , to the intermediary for moderate b , and does not delegate for high b .
- The figure makes clear that introducing an intermediary expands the range in which some form of delegation is valuable.

Economic message. The figure is the paper’s organizational-design summary. Authority should be closer to the informed agent when preference conflict is small, moved to an intermediary when conflict is moderate, and retained by the principal when conflict is too severe.

Detailed interpretation. The intermediary region is central. It shows that organizations may delegate to middle managers not because middle managers know more than frontline agents, but because their preferences are better aligned with top management. The intermediary acts as a commitment device that constrains the informed agent while preserving more information than direct principal-agent communication would.

7 Short summary

- The paper studies delegation as an alternative to strategic communication.
- The agent observes the state and has a systematic bias b relative to the principal.
- The principal can retain authority and communicate, or delegate formal decision rights.
- Delegation uses the agent’s information perfectly but implements the agent’s biased ideal action.
- Communication preserves control but generates noisy equilibrium partitions.
- Under a uniform prior, delegation dominates communication whenever communication is informative.
- For any smooth prior, delegation dominates for sufficiently small bias.
- Communication can dominate only when the principal’s prior is sufficiently precise relative to the bias.
- Delegation to an intermediary can be optimal for intermediate preference conflicts.
- Veto power can hurt because it adds variance to the bias unless the conflict is extreme.
- The main lesson is that authority is a way to preserve soft information when communication is strategically distorted.

8 Conclusion

8.1 Core conclusion

Dessein shows that delegation can be a superior way to use dispersed information inside organizations. When local managers have soft information and predictable biases, communication with superiors becomes strategic and noisy. Delegating authority sacrifices control but avoids the endogenous loss of information created by cheap talk.

8.2 Economic intuition

The intuition is that keeping authority changes what the informed agent is willing to say. The principal will correct for the agent’s bias, and the agent anticipates this correction. As a result, the agent cannot credibly reveal precise information. Delegation avoids this communication distortion by letting the agent act directly, but the resulting action is biased. The optimal authority structure balances these two errors.

8.3 Organizational implications

The theory explains why firms decentralize decisions even when local managers are biased. It also explains why middle managers and intermediaries can be useful. They need not possess the best information; they may simply have preferences that make their discretion safer than full delegation to frontline agents while still preserving more information than centralized communication.

8.4 Policy and management takeaway

Managers should not ask only whether subordinates are biased. They should ask whether the bias is large relative to the subordinate's informational advantage. If local information is very valuable and the bias is moderate, delegation can dominate monitoring and communication. If the bias is extreme or the superior already has a precise prior, central control may be better.

8.5 Limitations

The model abstracts from monetary incentives, endogenous information acquisition, multidimensional decisions, dynamic reputational concerns, and hard evidence. These are important in real organizations. But the abstraction is useful because it isolates a pure informational reason for decentralization.

8.6 Takeaway

The paper's lasting contribution is to show that decentralization is not merely a concession to local expertise or a response to incentive problems. It is a way to protect information from being destroyed by strategic communication.